Abdominal Transplant

Pre-CA1 education day
C Ryan King
10/24/2026

I am Ryan King I'm one of the transplant faculty. Last year I only used about half the time, so this year in addition to talking about real anesthesia preparation we will also have ITE focused keyword stuff. This lecture is about 2 surgeries: one that you are likely to do several times as a CA-1 (we have more than 300 kidney transplants a year and frequently assign first years) and one that you are reasonably unlikely to do (we do about 175 liver transplants a year but almost exclusively with senior residents).

Please listen to the surgical staff

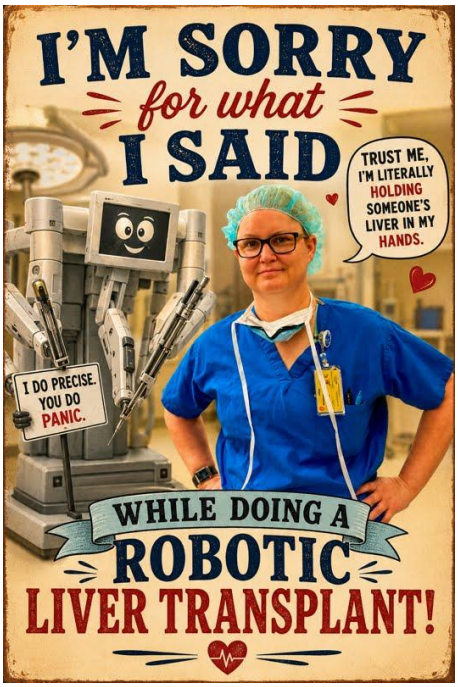
Meranda has been first assist for ~ 1500 liver transplants and many more kidneys.

She knows room layout and what they need.

Abdominal muscles recover from NMB faster than adductor pollicis, and ToF stickers come off.

She is interested in faster turnovers. Please cooperate. To facilitate them, ask for breaks.

You are entitled to work on the procedures for your cases, but after some trouble please call for support.



The surgical staff are generally correct when they say “we usually do X”. If you have specific concerns or reasons to think that something is unusual about the patient, please have a discussion.

Let’s get the easy stuff out of the way

Kidney transplantation is a great CA-1 case.

Usually *don’t* go through CPAP clinic, but have an extensive workup. Review:

- Cardiac function and ischemic testing (can be quite old!)
  - Very high rate of CAD
- Dialysis schedule, access, volume take off if today
- Urine production
- RBC antibodies (common) -> anticipated crossmatch time
- Functional status
- Focused for other systems

With a few caveats, kidney transplants are good cases early in your career.
For historical reasons, kidney transplant recipients don't go through our standard preoperative Clinic, but transplant Nephrology has a standardized evaluation which includes essentially everything that we would be interested in. the only caveat to that is that they are not always great about updating that data at a regular Cadence so for example they will have done some sort of ischemic evaluation on essentially everybody although it may have been a number of years. However, most of these patients will have seen us for some other surgery, such as dialysis access. we have a moderate number of patients who come up for kidney transplantation who have never been on dialysis but who have ckd5. these are not uncommonly patients who have identified a living donor. otherwise however a fair amount of what you will talk to the patient and pre-op about centers on their kidney disease such as when they last received dialysis if they use hemodialysis or peritoneal dialysis, any fistolas or arteriovenous grafts that have been created, and what their overall volume status is. often, these patients will have a good idea of whether or not they are volume up having played this game many times. Some patients coming up for kidney transplant make a small amount of urine or even a normal amount of urine if they are in the high output stage of kidney failure. that is important to assess both for tolerance of the urinary catheter placed in the OR and to understand what their pharmacokinetics are like. even a small amount of residual renal function is important as it can make the difference between clearing medications over a somewhat elevated amount of time Or over a very long. time, consider the example of sugammadex which is purely really eliminated

Because of the anemia induced by end-stage renal disease, many of these patients will have been transfused multiple times and will have anti RBC antibodies. typically one or two units of blood are prepared for this reason even though significant blood loss during the transplant itself is quite uncommon

Kidney transplant

- A normal GA
  - Roc / vec are *fine.* Sugammadex is fine (maybe slightly slower).
  - Sevo doesn't matter
  - Standard ESRD drug considerations
  - Ephedrine vs others unclear
  - Renal impact on hydromorphone is negligible
- Lab assessment (potassium, hgb)
- *Assess for moderate* fluid expansion
- Vascular access: try hard for a PIV

You can then expect a pretty normal general anesthetic as your plan. because of their biliary elimination you can expect that roc and vec will be eliminated a bit slower but still usable. some patients will have moderately elevated potassium, but they often have had elevated potassium many times and tolerate it reasonably well if they need sux. You may see concerns about sevo in older sources which are completely unfounded. although I personally believe that if adren is better for the reperfused kidney at the end of the case I have to admit that there is no high quality data showing an improvement in outcomes with ephedrine versus norepinephrine versus phenylephrine has a primary vasopressor.

the transplant team will have sent laboratory values that day or ordered a point of care that morning. it is uncommon to need to cancel somebody due to elevated potassium although it does happen occasionally that a patient needs dialysis or they are a suitable for surgery. the surgery team is going to want to avoid transfusions preoperatively if possible to minimize the potential for additional immune sensitization , but if the patient is symptomatically anemic then they will probably need it to tolerate anesthesia.

There is not a rigid fluid expansion target. as mentioned before some patients will come in having just completed dialysis and quite dry and other patients may not have had dialysis in several days and be somewhat volume overloaded. you have to look at their echocardiogram and their current status to assess whether or not they're going to tolerate any additional fluids.

We will try to have two decent sized peripheral IVs, so that they are continuous infusion of immunosuppressant can be switched to another IV if one stops working. we no longer place routine arterial access (and I would suggest that you think really hard about avoiding it), and if we can struggle through the case it is now preferred for Interventional Radiology to place a central line post-op because they have a different brand that is considered acceptable for the patient to take home for ongoing blood draws the first week or so. Although I love putting big peripheral IVs in anyone who has veins, in patients with end-stage real disease it is important to consider both vein preservation for future dialysis access targets if this kidney stops working someday and that these patients are pretty high risk to become peripherally exhausted before they leave the hospital. as a result, I try to avoid the cephalic vein at the wrist and the basilic vein, and don't go for anything larger than an 18 gauge.

Kidney transplant

Things that other people manage

- Immunosuppressants
- Reperfusion cocktail
- Postop dialysis / dialysis access
- Urine output!

- NAC, fenoldopam, mannitol all negative or inconclusive data
- Wake up, go to PACU

although there are several options for immunosuppression, there are only two that are routinely used here. the patients will get a weight based methylprednisolone Plus either basilixamab or antithymocyte globulin. They are both somewhat expensive products, so the Nephrology team will figure it out. Similarly, the surgeons will want arbitrary quantities of furosemide and a small vial of mannitol to promote initial urine output. They'll tell you what they want.

You don't need to worry about postop dialysis, because they will have a watch and wait approach for renal recovery.

The foley in a kidney transplant is a special setup such that you won't be able to measure urine output.

There are several drugs that you will see in the literature that do not have enough data to make them routinely used here.

Dopamine

Increases HR, CO. Mixed vaso-dilation and -constriction.

In *healthy* research participants *increases* renal blood flow (very high quality data)

In *kidney donors* improves post-transplant function (very high quality data)

In *kidney recipients* intraop and postop dopamine does not improve function and likely worsens complications (moderate quality data)

- https://pubmed.ncbi.nlm.nih.gov/8424517/
- https://pubmed.ncbi.nlm.nih.gov/28072739/
- https://pubmed.ncbi.nlm.nih.gov/10617237/
- https://pubmed.ncbi.nlm.nih.gov/20005360/
- https://pubmed.ncbi.nlm.nih.gov/15940046/

In patients with vasodilated or cardiogenic shock, dopamine does not improve or worsens renal function and increases tachyarrhythmias vs norepi (high quality)

I don't get this question as much anymore, but since it does show up on tests sometimes you do not want to give dopamine to kidney transplant recipients.

Liver transplant: Be not afraid

Your attending will help you. Questions are expected.

Your tech knows the defaults for set up.

There is an *exhaustive* protocol.

Some procedures will take longer … that's ok.

Most of our liver transplant recipients *aren't that sick*

You don't need to already know how to use the ROTEM.

It's ok to be nervous on your 1st / 2nd rodeo.

You don't need to be there an hour ahead of time. I go see the patient at T - 15 minutes.

There *almost always* is a pre-evaluation or consult note from anesthesia. Review:

> T&S (antibodies?); ECG (QTc); last EGD (banded in last month?); Hgb, plt, Cr, Na, K, INR, fibrinogen; TTE (PASP).

You don't need a bunch of drips. NE and a tube.

You don't need a pre-procedure arterial line.

You don't need to order blood to the room *before getting to the room.*

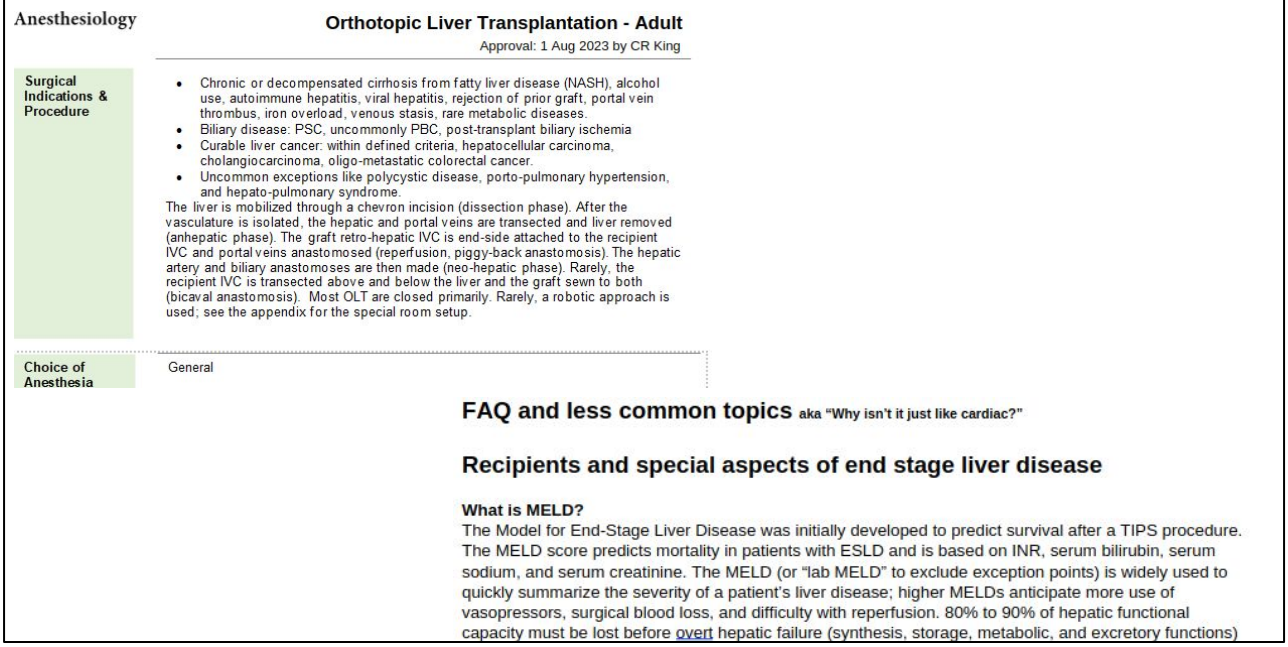
A main thing that I want you to take away is that you don't need to be afraid of this. liver transplantation is obviously a major case, and it is one of the bigger cases that does not explicitly require people to have done their cardiac rotation. there are multiple lines of safety to make sure that we do this in a highly reliable way.

Also as a ca-1, there are pretty strong guardrails to prevent you from getting stuck in a case that you are not ready for. most of the time, c a ones will only work on a liver transplant after they have done their cardiac rotation and are there for more familiar with some of the procedures, blood management, vasopressor management issues that are likely to come up. it is worth pointing out though that the pathophysiology is completely different from cardiac. it is possible for a ca-1 to be assigned into a liver transplant room after they have done some other general or rotations if there is enough attending staff to make them one-on-one with you.

as long as whoever you are working with knows where you are in your career then no one will be really disappointed or surprised.

we have always done a pretty exhaustive workup and every patient listed for liver transplant has been reviewed by one of the transplant anesthesia attendings with a note highlighting things that we think you need to know.

The protocol is pretty good



Anesthesiology

Orthotopic Liver Transplantation - Adult

Approval: 1 Aug 2023 by CR King

Surgical Indications & Procedure

- Chronic or decompensated cirrhosis from fatty liver disease (NASH), alcohol use, autoimmune hepatitis, viral hepatitis, rejection of prior graft, portal vein thrombus, iron overload, venous stasis, rare metabolic diseases.
- Biliary disease: PSC, uncommonly PBC, post-transplant biliary ischemia
- Curable liver cancer: within defined criteria, hepatocellular carcinoma, cholangiocarcinoma, oligo-metastatic colorectal cancer.
- Uncommon exceptions like polycystic disease, porto-pulmonary hypertension, and hepato-pulmonary syndrome.

The liver is mobilized through a chevron incision (dissection phase). After the vasculature is isolated, the hepatic and portal veins are transected and liver removed (anhepatic phase). The graft retro-hepatic IVC is end-side attached to the recipient IVC and portal veins anastomosed (reperfusion, piggy-back anastomosis). The hepatic artery and biliary anastomoses are then made (neo-hepatic phase). Rarely, the recipient IVC is transected above and below the liver and the graft sewn to both (bicaval anastomosis). Most OLT are closed primarily. Rarely, a robotic approach is used; see the appendix for the special room setup.

Choice of Anesthesia

General

FAQ and less common topics aka "Why isn't it just like cardiac?"

Recipients and special aspects of end stage liver disease

What is MELD?

The Model for End-Stage Liver Disease was initially developed to predict survival after a TIPS procedure. The MELD score predicts mortality in patients with ESLD and is based on INR, serum bilirubin, serum sodium, and serum creatinine. The MELD (or "lab MELD" to exclude exception points) is widely used to quickly summarize the severity of a patient's liver disease; higher MELDs anticipate more use of vasopressors, surgical blood loss, and difficulty with reperfusion. 80% to 90% of hepatic functional capacity must be lost before overt hepatic failure (synthesis, storage, metabolic, and excretory functions)

The protocol from 2023 is fine there is a second version from last year that is organized slightly differently but has essentially the same content. please do not look at the really old thing labeled resident manual.

There's also a transplant fellow manual which goes into exhaustive detail on many issues if you are curious or want to see the underlying data

OLT indications

Acute liver failure, primary graft non-function

ESLD with biochemical decompensation (MELD > 15)

ESLD with severe, reversible complications

- Hepatopulmonary syndrome, porto-pulmonary hypertension

Cholangiocarcinoma, hepatocellular carcinoma, resectable colorectal cancer

Cystic fibrosis

Hepatic artery thrombosis

Familial amyloid, several metabolic diseases, primary hyperoxaluria

Polycystic liver disease (pain) (nonstandard)

TIPS refractory ascites (nonstandard)

So far as interns you have probably taken care of patients who have had liver transplants in the SICU or who need liver transplants on the medicine service. unfortunately, many of the patients with end-stage liver disease who you have taken care of will simply not be eligible because they have a disease that is not curable. often that as alcohol use disorder, although sometimes it is cancer that is no longer curable. you have probably seen the m e l d score used to discuss the severity of someone's liver disease, and in patients who have primarily a synthetic dysfunction disorder that is still used as a summary. A MELD 15 = net survival benefit, but you will see many patients with other complications of cirrhosis or related diseases who are eligible candidates for transplantation.

This is a huge spectrum of severity of illness!

Easy cases

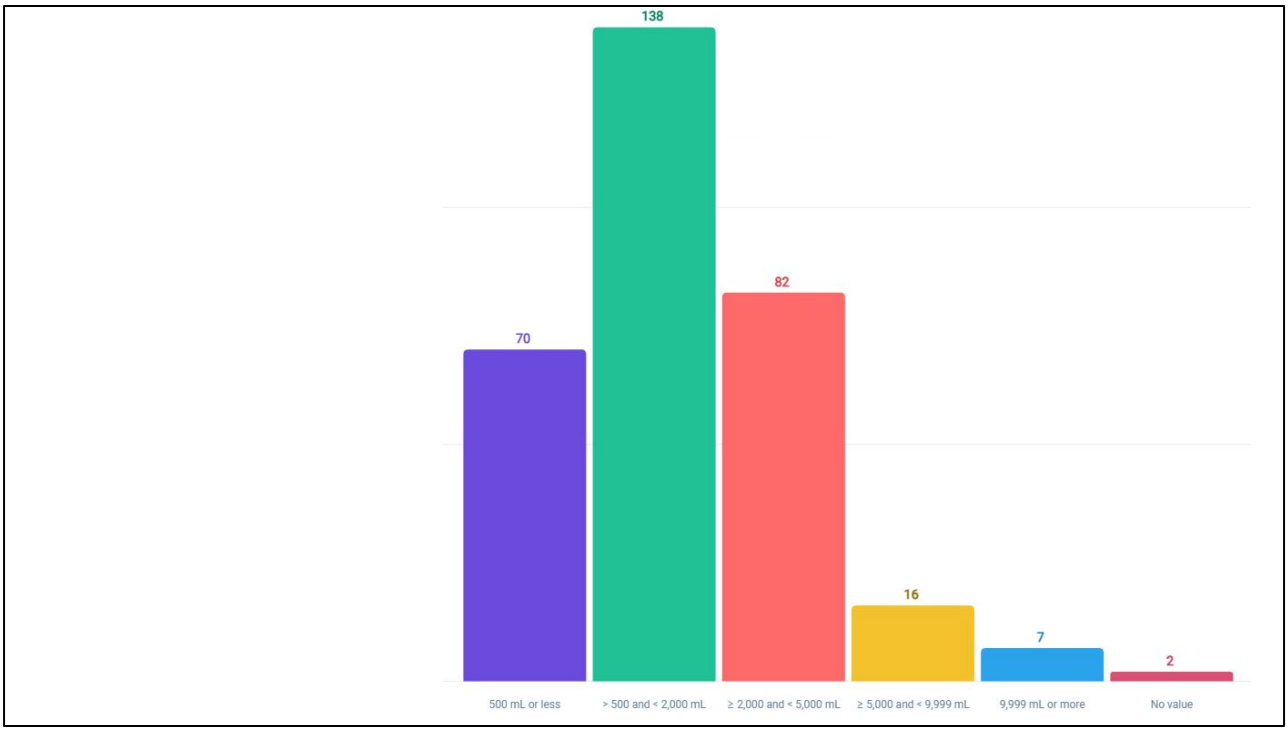
- Cancer (HCC, cholangio, CRC) - often normal biochemical disease!
- Moderate liver disease *and* need for a kidney transplant

Kind of hard

- ESLD with moderate biochemical disease and symptoms / complications
- ESLD with moderately bad complications (porto-pulmonary, hepato-renal)
- Retransplant for ischemic cholangiopathy

Very bad

- Acute liver failure (acetaminophen)
- Primary graft non function



This is the distribution of EBL As you can see, a fair number of these are chip-shots with 500 mL or less of EBL, and it is uncommon for us to go over 1-2 blood volumes. In the middle it is often a lot of work keeping up with intermittent blood loss, but not exsanguinating with the belmont maxed out like you might see in some trauma cases.

These high-loss cases are surprisingly predictable, so you won't be caught unawares.

Allocation

Liver: sickest first

Acute failure: 1a or MELD=40

MELD [2002]: log(bili) + log(INR) + log(Cr)

MELDNa [2016]: MELD + (137 - Na) + M*(137-Na)

MELD3.0 [2023]: + female + albumin + albumin:[Na + log(Cr)]

Exceptions: MMAT - 3

Kidney: Time on dialysis or waitlist

Longevity matching

Exceptions for donors, kids, difficult xmatch

Allocation Circles

Replaced donor service areas

Sequentially offer a donor to recipients in

[1a , 37+ , 33+ , 29+ , 15+] in radii of

[150, 250, 500] nm.

DCD or donor age > 70: just [150, 250, 500].

Peds first within each "bucket"

Order by MELD within "bucket"

Variances for AK, HI, PR

Each Transplant Center has a responsibility to evaluate potential transplant recipients and submit them into the national league organized list. unlike kidney transplant, which is largely based on time, for liver transplant wait list is something of a misnomer, because Acuity is based on expected survival without a transplant.
I mentioned that an m e l d score of greater than 15 is where there is usually a survival benefit, but most patients with liver failure will be substantially higher. we allocate livers in a sickest First fashion mostly based on their laboratory values, although there are many exception point criteria to increase someone's position on the list that you do not need to know at this stage. Sickest first != net benefit, but almost
Meld originally for TIPS survival (small cohorts!). Adopted to replace CP because objective. More and more nonlinear / ML models. Oddly, require validation in non-transplant candidates
donor organs come largely from our local area but can be sourced from up to 500 nautical miles away.

What actually happens

You don't need to be there an hour ahead of time. I go see the patient at T - 15 minutes.

There *almost always* is a pre-evaluation or consult note from anesthesia. Review:

> T&S (antibodies?); ECG (QTc); last EGD (banded in last month?); Hgb, plt, Cr, Na, K, INR, fibrinogen; TTE (PASP).

You don't need a bunch of drips. NE and a tube.

You don't need a pre-procedure arterial line.

You don't need to order blood to the room *before getting to the room.*

Room set up:

Induction: Prop+roc+lido

Available: Vaso, insulin, "liver box," zosyn

NE infusion

McGrath, normal sized ETT

Quad lumen CVC

Arterial/CVP pressure bag

A-line: tegaderm, needle, 4x4, prep stick, gloves

RIC: tegaderm, 20g, RIC, prep, gloves, towels

NGT

US, TEE (one is permanently in 225)

One thing that I think stresses people out about their first couple of liver transplants is the idea that they need to be here really early and to get a bunch of stuff set up.
however, because there is a relatively High cancellation rate for transplants when they take the organ out of the donor and find that it is not suitable, we actually don't want you to set up a bunch of things. the expectation is that we will get to the or and help each other.
I personally show up 30 to 15 minutes before a transplant so that I can sit down and talk with them and see if there is anything that needs to be addressed. when I was the fellow and did around 80 of these in a year, I would show up around 30 minutes ahead of time get set up over the course of around 15 to 20 minutes, and go talk with the patient before rolling back. it is uncommon that patience need a pre procedure arterial line.

there are some specifics listed here, but unlike cardiac and lung transplant, our liver transplant recipients almost all go through CPAP or they have an inpatient evaluation by one of the transplant attendings which shows up as a consult note.
these patients will often have millions of dollars of lab work that has seemingly been done but there's actually a relatively short list of things that I would expect you to review the day of service, most of them are in the anesthesia labs summary. the most important one to check usually is if they have developed antibodies on their typing screen.

you also definitely should not expect to set up a million drips. almost everyone will get induced with just a vasopressor and possibly a carrier. the blood ordering is

almost completely standardized, and the only thing that trips people up is occasionally bringing platelets to the room that we intended to stay downstairs, unlike in cardiac surgery.

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NGT

Tech: Arterial/CVP bag, belmont, Quad lumen

US, TEE, twitch view, BIS (permanently in 225)

As I mentioned, the room setup itself is not at all involved and should not take you very long. 98% of the time you need the completely standard set of induction medications norepinephrine set up and then a handful of additional medicines that the pharmacy will get out when you say that you're doing a liver transplant. completely normal intubation equipment and then a couple of supplies for an arterial line and a rapid infusion catheter if you want to do one of those. the tech will set up a lot of standard equipment. you can't go wrong by just asking them to open a quad lumen, and then if you need to double stick to patient and we can just hand you a cordis. there is a fair amount of equipment that lives permanently in 225 so there's no need for you to go hunting for it.

The pathophys is very different from cardiac, toolkit similar

Cardiac	Liver
Low CO due to valvular or cardiomyopathy -> low BP -> vasopressors, often inotropes -> central line, arterial line	High CO with severe vasodilation -> low BP -> vasopressors, rare inotropes -> central line, arterial line
Cardiac dysfunction, PA-HTN, air -> TEE	Volume loss and vasodilation, rare air -> TEE
Significant blood loss, arms tucked -> cordis	Significant blood loss, arms out -> cordis or RIC
Pump coagulopathy, heparin effects -> ROTEM	Intrinsic coagulopathy -> ROTEM
Consumption, pump adherence, small arterial anastomosis -> frequent, multiple platelet transfusions plus antiplatelets	Splenic sequestration, marrow suppression, consumption, small arterial anastomosis -> sad about it
Low CO -> high CVP, treat heart function	Fixed High CO + trend CVP -> +- volume
Autologous “good blood” before pump	Autologous “bad blood” before surgery

During the case

Dissection: communicate closely, labs hourly

Anhepatic: communicate closely, give steroids, labs when out of ice

Neohepatic: communicate closely

Post-reperfusion: communicate closely, labs as needed, usually hourly

Talk about:

All the time:

Volume goals

Coagulation and platelet goals

Acid-base goals

Na, Ca, K

Anhepatic: everything

Post-reperfusion: extubation

A liver transplant can be divided into four phasers the first phase is when they are chipping the liver out called the dissection. the second phase is when they take that liver out and get ready to put the new one in. the third phase is the first 15 minutes or so after the new one is in. The final phase is the rest of the case where they get the artery and the bile duct hooked up and close the patient. the major expectation during the entire case is that you are going to communicate closely with the attending who will either have no other room or something minor that they don't need to pay attention to. There are a lot of details about what we're doing in transplant, but most of the time if you are talking to me frequently about the volume resuscitation, the coagulation acid base and electrolyte issues, that is going to be most of it. we end up Excavating around half of these patients, and it is often difficult to tell who is going to be ready to extubate until the final hour or so, so unless something has gone very difficult you can just tee everyone up as if they were going to extubate and then be sad about it if they're not ready.

Portal hypertension is the critical concept

ΔPortal pressure ~ outflow pressure + (portal flow × resistance)	
Inflow	Pathologic splanchnic arterial dilation increases. Increases with CO
Resistance	Cirrhosis: hepatic resistance. Collaterals: parallel outflow, nonlinear capacitance and pressure dependent compliance.
Outflow	CVP = afterload, high inhibits drainage.

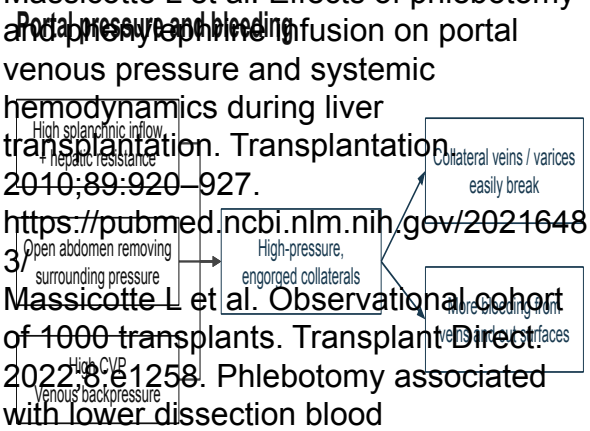
Sarin SK, Pamecha V, Sharma BC, Kapoor D. Portal hypertension and its surgical management in the Indian subcontinent. Liver Transpl. 2011;17:836–848.

Sarin-Pamecha et al. Lack of a correlation between portal vein flow and pressure. Liver Transpl 2011;17:836–848. Prospective measurements in 81 transplants demonstrate that flow and pressure require separate assessment.

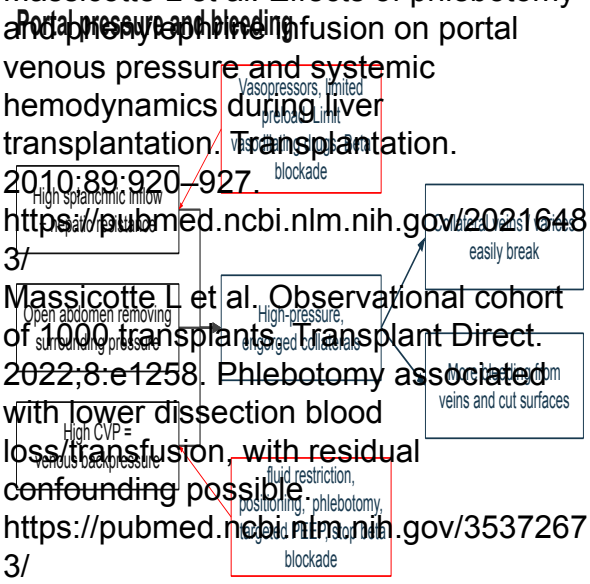
https://doi.org/10.1002/lt.22295

Massicotte L et al. Effects of phlebotomy and phenylephrine infusion on portal venous pressure and systemic hemodynamics during liver transplantation. Transplantation. 2010;89:920–927. https://pubmed.ncbi.nlm.nih.gov/20216483/

Massicotte L et al. Observational cohort of 1000 transplants. Transplant Direct. 2022;8:e1258. Phlebotomy associated with lower dissection blood loss/transfusion, with residual confounding possible. https://pubmed.ncbi.nlm.nih.gov/35372673/



Sainz-Barriga M et al. Liver Transpl. 2011;17:836–848. Prospective flow and pressure measurements, including variable CVP transmission. https://doi.org/10.1002/lt.22295



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Sainz-Barriga M et al. Liver Transpl. 2011;17:836–848. Prospective flow and pressure measurements, including variable CVP transmission. https://doi.org/10.1002/lt.22295

Nadim MK et al. Acute kidney injury in patients with cirrhosis: ADQI and ICA joint multidisciplinary consensus. J Hepatol. 2024. https://doi.org/10.1016/j.jhep.2024.03.031 Consensus source: https://pmc.ncbi.nlm.nih.gov/articles/PMC11193657/

Splanchnic vasodilation and HRS

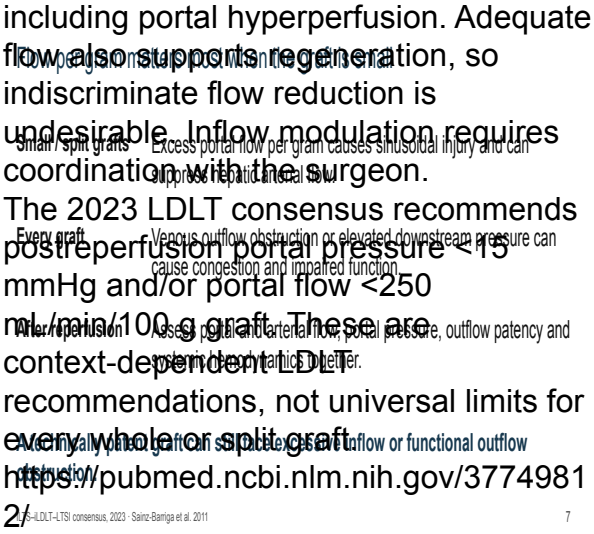
Arterial underfilling	A dilated splanchnic bed and excess total body fluid can coexist with low MAP and low renal artery perfusion.
Renal response	Neurohumoral activation causes renal vasoconstriction and sodium retention.
Intraoperative implication	With adequate CO and vasoplegia, more flow may feed the splanchnic bed. Restoring vascular tone may better support pressure. Vasopressin + analogues *increase* renal perfusion.
Afterload management	High CVP reduces the perfusion gradient and promotes interstitial congestion.

A high cardiac output does not guarantee adequate renal perfusion.

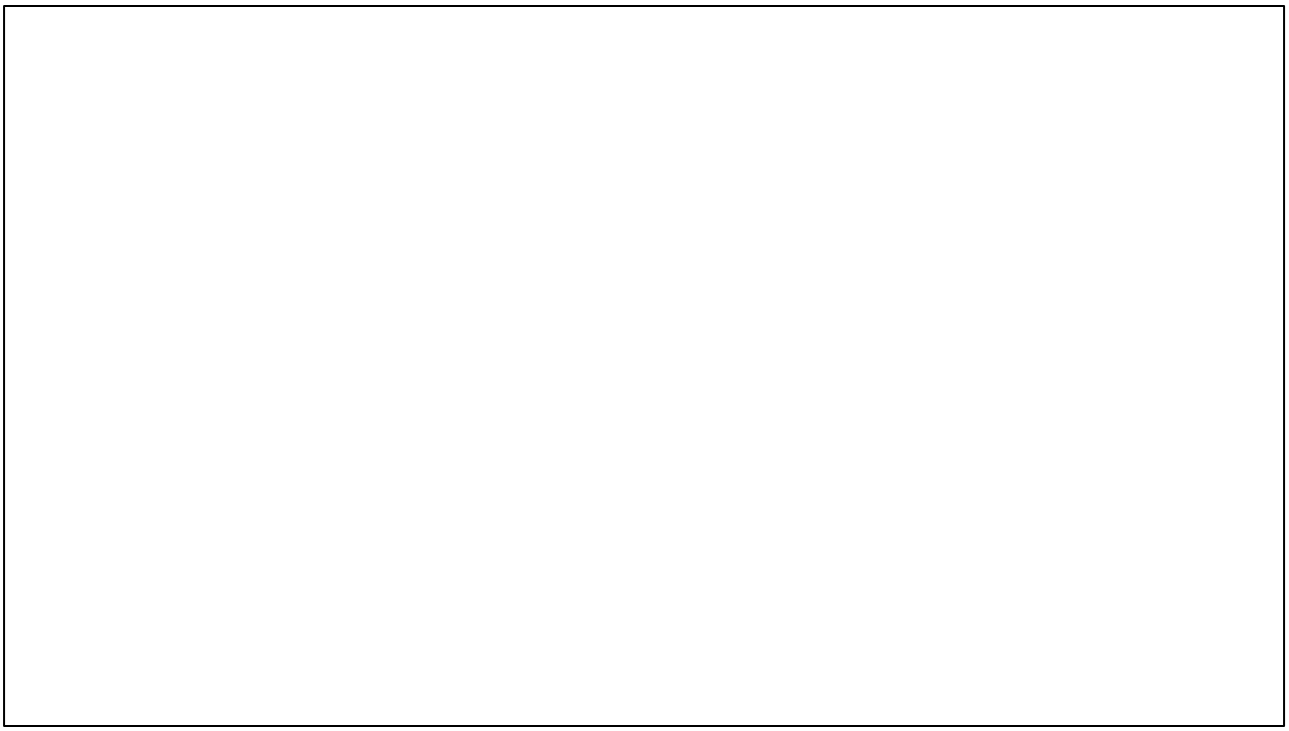
Small-for-size syndrome reflects graft size/quality and recipient demands, including portal hyperperfusion. Adequate flow also supports regeneration, so indiscriminate flow reduction is undesirable. Inflow modulation requires coordination with the surgeon.

The 2023 LDLT consensus recommends postreperfusion portal pressure <15 mmHg and/or portal flow <250 mL/min/100 g graft. These are context-dependent LDLT recommendations, not universal limits for every whole or split graft.

https://pubmed.ncbi.nlm.nih.gov/37749812/



Preventing SFSS consensus emphasizes adequate outflow reconstruction and portal inflow modulation. https://pubmed.ncbi.nlm.nih.gov/37635285/
Sainz-Barriga et al. 2011: portal flow and pressure did not correlate reliably during LT. Assess both alongside systemic hemodynamics. https://doi.org/10.1002/lt.22295
Impact of functional hepatic venous outflow obstruction on perioperative outcome after living-donor liver transplant. Observational study associates higher CVP with poorer outcomes, without establishing a universally harmful threshold. https://pubmed.ncbi.nlm.nih.gov/29363416/



ABA Stuff

4. Gastrointestinal/Hepatic Systems

b. Clinical Science

2) Hepatic Disease

a) Preoperative Laboratory Assessment

b) Anesthesia Choice (Hepatocellular Disease, Ascites, Portal Hypertension)

c) Postoperative Hepatic Dysfunction, Hepatic Failure, Hepatorenal Syndrome

d) Hepatic Transplantation

5. Renal and Urinary Systems/Electrolyte Balance: Clinical Science

a. Renal Disease

4) Anesthetic Management in Renal Transplantation

One way to come at this topic is to ask what do you need to know for the boards. that is probably the easiest question to answer: almost nothing. what you see on this slide are the only topics on the ABA content outline relating to abdominal transplant. you are much more likely to be asked questions about hepatic physiology or end-stage liver disease for non-transplant surgeries. one thing to remember is that many anesthesia residents are at programs that do know or very few liver transplants, and will never see one. Many of the practices in liver transplant anesthesia have never been subjected to rigorous trials, and vary center to Center. which also makes them bad topics for a national standardized exam.

you are obviously responsible for understanding how to do kidney transplants, but there is relatively little that makes a good question about kidney transplants.

ABA Stuff

2. Organ Donors: Pathophysiology and Clinical Management

a. Circulatory

b. Endocrine

c. Fluid and Electrolyte Management

d. Brain Death Criteria

It isn't quite in what we would think about as the same topic, but you are reasonably likely to be asked questions about the management of brain dead or severely brain injured individuals who are organ donor candidates. again, this is the content outline material which is underneath the critical care section of the outline.

Transplant topics

General OLT criteria

MELD, CPT -> waitlist

Neat but low-yield topics

Acute liver failure - intracranial pressure

Transfusion management in ESLD

Temperature management

Anti-rejection meds

Electrolyte / acid management

Reperfusion management

Sodium management

Related topics

TEG / ROTEM

MTP

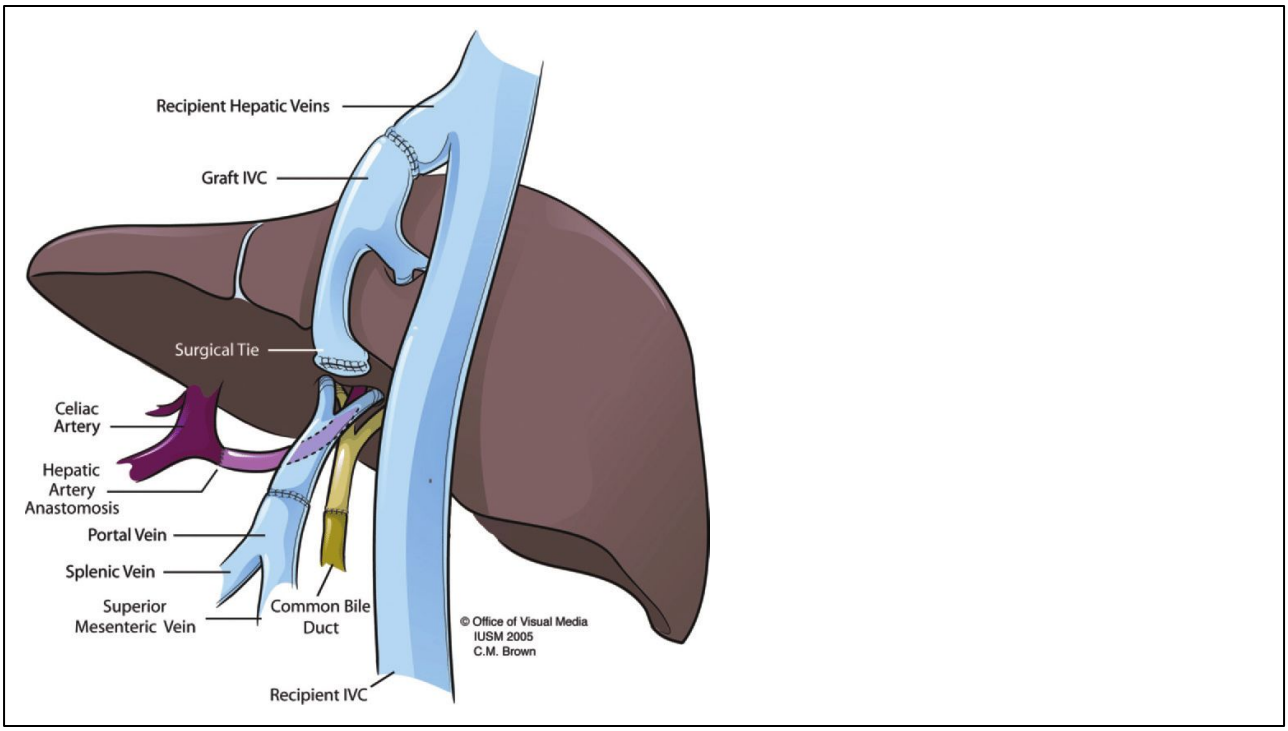
TEE interpretation

ESLD Physiology

"Knocking out the boards" has a nice review of ESLD topics

- Cardiac
- Electrolytes
- Renal function / hepatorenal type 1 vs 2
- Portal hypertension and ascites
- Hepatopulmonary
- Portopulmonary HTN
- Coagulation

So we end up with this division of topics where there are a tiny number of things to know that are both specifically about liver transplantation and fair game for your exam. there are lots of topics that come up in liver transplantation that are on your exam including a fair amount of end-stage liver disease physiology, which are reasonably high yield to review, But are not really specific to transplant. there are also topics that are kind of interesting and important to the actual management of livers, but just very unlikely to show up on your exams


Recipient Hepatic Veins
Graft IVC
Surgical Tie
Celiac Artery
Hepatic Artery Anastomosis
Portal Vein
Splenic Vein
Superior Mesenteric Vein
Common Bile Duct
Recipient IVC
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